

Notes on Renewable Energy

Solar Energy

Solar energy is generated by converting sunlight into electricity, either directly using photovoltaic (PV) panels or indirectly using concentrated solar power. Photovoltaic cells are made of semiconductor materials, typically silicon, which absorb photons and release electrons, creating an electric current. Solar panel efficiency has improved significantly over the past decade, with commercial panels now averaging 20-22% efficiency, up from around 15% in 2010. The main advantages of solar energy are its abundance, low operating costs after installation, and minimal environmental impact during operation. The primary drawback is intermittency, since solar panels only generate electricity when sunlight is available, requiring battery storage or grid backup for consistent power.

Wind Energy

Wind energy is captured using turbines that convert kinetic energy from moving air into mechanical power, which is then converted into electricity by a generator. Modern wind turbines typically have blades between 40 and 90 meters long, mounted on towers reaching heights of over 100 meters to access stronger, more consistent winds. Wind farms can be located onshore or offshore, with offshore wind farms generally producing more consistent power due to stronger and steadier ocean winds, though they are more expensive to construct and maintain. Denmark generates over 50% of its electricity from wind power, making it one of the global leaders in wind energy adoption.

Hydroelectric Power

Hydroelectric power generates electricity by using flowing or falling water to spin turbines connected to generators. It is the most widely used form of renewable energy globally, accounting for roughly 16% of total electricity generation worldwide. Large-scale hydroelectric dams, such as the Three Gorges Dam in China, can generate over 20,000 megawatts of power. Hydroelectric power is highly reliable and can be quickly ramped up or down to match electricity demand, making it valuable for grid stability. However, large dam projects can have significant environmental and social impacts, including displacement of communities and disruption of river ecosystems.

Energy Storage Challenges

A major challenge across all renewable energy sources is intermittency, since solar and wind power generation depends on weather conditions rather than electricity demand. Battery storage technology, particularly lithium-ion batteries, has become the leading solution for storing excess renewable energy for later use. Grid-scale battery storage costs have fallen by approximately 90% since 2010, making large battery installations increasingly economically viable. Other storage approaches include pumped hydro storage, which uses excess electricity to pump water uphill for later release through turbines, and hydrogen storage, which converts excess electricity into hydrogen gas that can be stored and later converted back to electricity.